

10 (a) (i) density $\times$ speed of wave (in the medium) B1 [1]

(ii) $\rho = (7.0 \times 10^6) / 4100$
 $= 1700 \text{ kg m}^{-3}$ A1 [1]

(b) (i) $I = I_T + I_R$ B1 [1]

(ii) 1. $\alpha = (0.1 \times 10^6)^2 / (3.1 \times 10^6)^2$ C1
 $= 0.001$ A1 [2]

2. $\alpha \approx 1$ A1 [1]

(c) *either* very little transmission at an air-skin boundary M1
 (almost) complete transmission at a gel-skin boundary M1
 when wave travels in or out of the body A1 [3]

or no gel, majority reflection (M1)
 with gel, little reflection (M1)
 when wave travels in or out of the body (A1)

11 (a) (i) unwanted random power / signal / energy B1 [1]